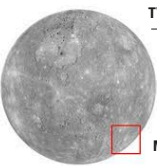


GEOLOGICAL FEATURES

More than 3 billion years ago, most of Mercury’s surface crust erupted as vast sheets of lava like the maria on the Moon but poorer in iron and richer in magnesium, silicon, and various volatile elements. Smaller areas, especially inside some craters, were flooded by lava more recently. There have also been explosive eruptions less than a billion years ago.

RENOIR REGION

Discovery Rupes



TYPE Ridge

AGE 2 billion years

LENGTH 310 miles (500 km)

MESSENGER MAP

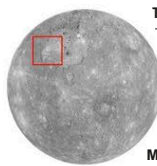
This escarpment (*rupes* in Latin), which in places is 1.2 miles (2 km) high, is younger than both the craters and the volcanic plains that it crosses. Discovery runs in a northeast-southwest direction and is 310 miles (500 km) in length. Many similar features, known as “lobate scarps,” have been mapped on Mercury. The planet’s slow cooling over the past 4 billion years has caused its radius to shrink by about 5 miles (7 km). As a result, adjacent tracts of the rock crust have been thrust over the adjacent terrain.



DISCOVERY RUPES CUTTING THROUGH IMPACT CRATERS

CALORIS BASIN FLOOR

Agwo Facula



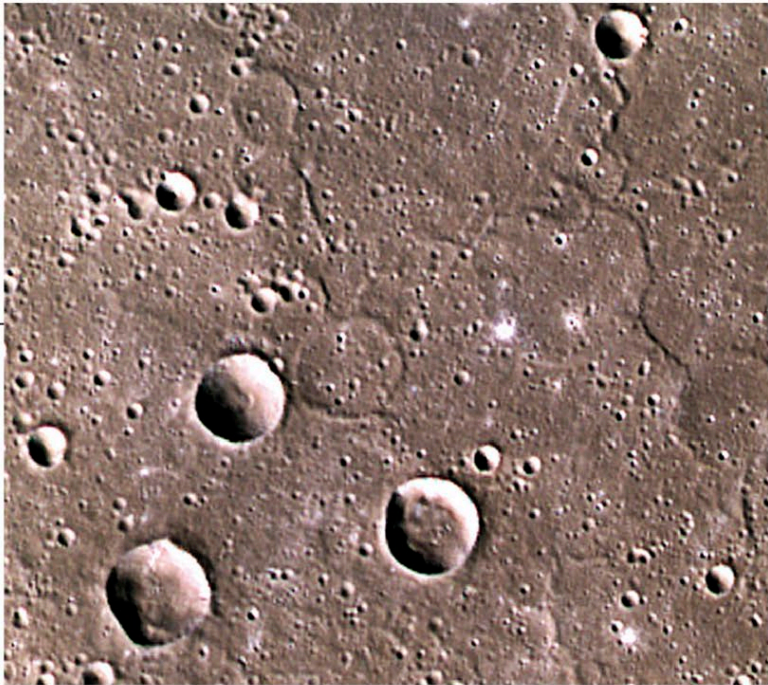
TYPE Explosive volcanism

AGE 2 billion years

DIAMETER 60 miles (100 km)

MESSENGER MAP

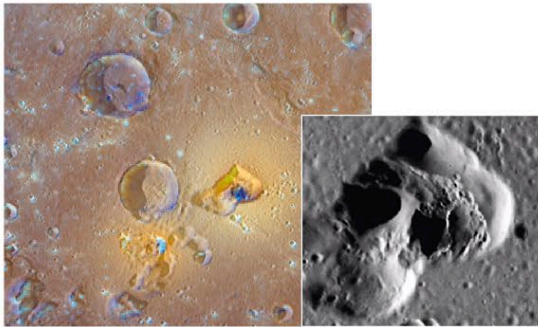
This orange-colored bright spot (*facula* in Latin) is one of nearly 200 examples on Mercury believed to be deposits resulting from explosive volcanic eruptions. These eruptions blasted out the irregular pits that can usually be seen near their midpoints. Each deposit gets fainter toward its outer edge, which blends into the background. The more powerful the explosion (or series of explosions), the bigger and brighter the deposit. The youngest such explosive eruption deposits on Mercury are probably less than a billion years old. Most eruptions on Mercury produced lava flows, but the less common explosive eruptions are significant



because they can happen only if there is enough gas to expand violently and cause an explosion when molten rock is erupted toward the surface.

PLAINS VOLCANISM

This image reveals faint circular traces of “ghost craters” that were completely flooded by lava, creating plains. More obvious craters were formed by later impacts into the lavas.

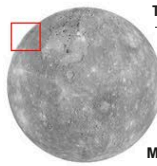


COMPOUND VENT

The 17-mile (27-km) long pit in the middle of Agwo Facula shows it to consist of several overlapping volcanic vents. The deepest exceeds nearly a mile (km) in depth.

RADITLADI REGION

Hollows



TYPE Surface degradation

AGE Under 100 million years

DIAMETER 100 yards (0.1 km)

MESSENGER MAP

The MESSENGER spacecraft revealed finer detail than previously seen on Mercury. One new discovery are features called hollows, which can occur singly but are more often in groups of dozens or hundreds. Hollows are individual patches of surface hundreds of feet (meters) wide, which have been stripped away to a depth of 33–66 ft (10–20 m), to leave steep-sided, flat-bottomed depressions.

Hollows must be young, because no impact craters on Mercury have been made inside them, and some hollows could still be forming today. Their shapes suggest gentle growth, in contrast to the larger and violently excavated volcanic vents

associated with faculae, but it is not known how or why they form. There is no air on Mercury, so it cannot be caused by scouring by wind. A volatile component of the surface material is probably being lost to space, perhaps because of Mercury’s searing noontime temperatures, which can exceed 800°F (400°C).

Comparing the surface composition inside and outside hollows in order to work out what is going on is one of the science goals of the BepiColombo mission.

DETAIL OF HOLLOWS

This detailed view in exaggerated color covering an area only 10 miles (16 km) wide shows steep-sided, flat-bottomed blue hollows on the red floor and peak-ring inside Raditladi crater.

